Kiran Muppana

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Experience

Software Engineer 2, Florida State University Oct 2024 – Present

- Engineered a scalable real-time energy management system using Java, SpringBoot, ReactJs and Kubernetes.
- Led critical frontend **performance** optimization, reducing the production bundle size by 70% and initial route size by 94% using React Dynamic Imports and code splitting to boost LCP and overall SEO.
- Migrated to Go with **gRPC** and Protocol Buffers, sustaining **1.8M TPS** per instance with zero scaling needs.
- Orchestrated horizontal sharding strategy for **PostgreSQL** clusters, improving database write performance by 2x under peak load.
- Worked on event **streaming** solution by building **Kafka** consumers and producers with multiple partitions for real-time event processing across four microservices.
- Configured **Jenkins** CI/CD builds and Helm charts, handling over **5M** daily API requests with 99.9% availability.
- Integrated third-party multi-cloud applications, implementing SSO and RBAC using **OAuth 2.0** and **Keycloak**.

Software Engineer, Florida State University Dec 2022 – Oct 2024

- Developed reactive, non-blocking APIs using **Spring WebFlux**, enabling high-throughput async service-to-service communication and reducing p95/p99 latency by 65%.
- Created autoscaling mechanisms on **AWS ECS** and **Fargate**, optimizing compute cost by 25%.
- Collaborated across 4 teams to standardize API contracts and enforce schema consistency with **OpenAPI 3.0**.
- Implemented Prometheus/Grafana **observability** with namespace isolation, reducing MTTR to under 15 mins.
- Deployed **Redis** for distributed caching and optimized key eviction policies, improving app responsiveness by 45%.
- Architected a workflow analyzing 150k documents to auto-generate work orders, eliminating manual steps and saving 30+ hours weekly.

Software Engineer, Amdocs Jul 2021 – Jul 2022

- Devloped RESTful AIOps microservice using **Java SpringBoot**, implementing layered traffic management (soft/hard rate limiting) and advanced failure protection via Circuit Breaking and Bulkhead isolation.
- Implemented a high-throughput data ingestion service using **AWS** Kinesis, Firehose, Glue, and Lambda, processing 2M+ events/sec and 10TB+ daily with scalable storage in S3.
- Optimized **SQL** queries and DB indexes, reducing report generation time from 3 minutes to under 30 seconds.
- Automated secure authentication using OAuth, JWT, and React Routes, ensuring seamless role-based navigation.
- Worked closely with QA to adopt **Test-Driven Development (TDD)** practices across the team.
- Built automated deployment pipelines with **Jenkins**, **Terraform** and integrated quality gates using **SonarQube**.
- Worked in Agile teams with regular code reviews, sprint planning, retro and cross-functional collaboration.

Technical Skills

Programming Languages: Java, Go, Python, JavaScript, SQL, Shell
Backend Frameworks: Spring Boot, WebFlux, Multi-Threading, gRPC, Gin, Kafka Streams
Frontend: React, Next.js, TypeScript, Tailwind CSS
Cloud/DevOps: AWS (ECS, Lambda, Kinesis), Kubernetes, Terraform, Docker, Jenkins, GitHub Actions
Databases: PostgreSQL, Oracle DB, MongoDB, Redis, Snowflake
Observability: Prometheus, Grafana, AppDynamics, ELK Stack, Jaeger, K6

Education

Florida State University, Tallahassee, FL – Master's in Computer Science (May 2024)
National Institute of Technology, Calicut, India – Bachelor's in Electronics & Communication (May 2021)

Certifications & Publications

Certifications: Oracle Certified Generative AI Professional | Oracle Certified Data Science Professional
Publication: *Transformer models for enhancing AttnGAN based text to image generation* – Elsevier